

Credit Risk Analytics & Securitisation Waterfall Analysis

ECL Modelling • Stress Testing • Tranche Analysis • Power BI Dashboard

Professional Project Report

Prepared for professional project submission

1. Executive Summary

This project presents an end-to-end credit risk analytics framework for a synthetic loan portfolio of 5,000 loans with total exposure of approximately ■231.45 million. The analysis evaluates portfolio credit risk using Probability of Default (PD), Loss Given Default (LGD), Exposure at Default (EAD), and Expected Credit Loss (ECL).

2. Project Objectives

The objectives are to analyze portfolio credit risk, calculate ECL using $PD \times LGD \times EAD$, segment risk by asset type, credit score, LTV, DPD and credit-risk stage, evaluate adverse scenarios through stress testing, demonstrate a prototype securitisation structure, analyze waterfall loss allocation, and present the results through an interactive Power BI dashboard.

3. Dataset & Portfolio Overview

The analytical dataset contains 5,000 synthetic loans and 17 variables. It is designed for demonstrating a reproducible credit-risk workflow rather than representing an actual lender portfolio.

4. ECL Methodology

Expected Credit Loss is calculated at loan level using $ECL = PD \times LGD \times EAD$. Loan-level ECL values are aggregated to portfolio-level metrics and analyzed across asset type, credit-risk stage, credit score, LTV and days past due.

5. Risk Analysis

Risk segmentation is performed across credit-score bands, LTV bands and DPD bands. Asset-type and stage-level analysis provides additional views of portfolio concentration, default risk and expected loss.

6. Stress Testing

Three analytical scenarios are used: Base (PD 1.00×, LGD 1.00×), Moderate Stress (PD 1.30×, LGD 1.10×), and Severe Stress (PD 1.70×, LGD 1.25×). These multipliers are illustrative assumptions used to assess sensitivity of expected losses under adverse conditions; they are not forecasts.

7. Securitisation & Tranche Analysis

A simplified prototype waterfall is used with Senior, Mezzanine and Equity tranches sized at 70%, 20% and 10% respectively. Portfolio losses are allocated sequentially from Equity to Mezzanine and then Senior. This is an analytical demonstration and does not represent legal transaction terms.

8. Power BI Dashboard

The Power BI reporting layer contains four pages: Credit Risk Dashboard; Stress Testing & Securitisation Analysis; Risk Analysis; and Portfolio & ECL Summary. Together, the pages provide portfolio exposure, ECL, default rates, stress scenarios, tranche structure, waterfall analysis and risk segmentation.

9. Key Findings

The portfolio contains 5,000 loans with approximately ■231.45 million of exposure and ■7.68 million of total ECL. Portfolio ECL is approximately 3.32% of exposure, with an overall default rate of 30.36%. Stage 3 has a higher ECL rate than Stage 1 and Stage 2. Stress scenarios increase expected losses as PD and LGD assumptions are increased. The waterfall demonstrates the role of subordinated tranches in absorbing losses before the Senior tranche under the simplified assumptions.

10. Limitations

The portfolio is synthetic; model assumptions are not externally validated; stress scenarios are illustrative; the securitisation waterfall is simplified and excludes legal, structural, servicing, liquidity and transaction-specific features; and the analysis should not be interpreted as lending, investment or transaction advice.

11. Conclusion

The project demonstrates a complete workflow from loan-level data preparation through ECL modelling, risk segmentation, stress testing, prototype securitisation waterfall analysis and business intelligence reporting. It combines Python-based analytics with Power BI to convert detailed portfolio data into decision-oriented risk reporting.

Portfolio Snapshot

Metric	Value
Number of loans	5,000
Total exposure (EAD)	■231.45 million
Total ECL	■7.68 million
Average ECL	■1,536.88
ECL / exposure	3.32%
Default rate	30.36%

Note: All portfolio values and analytical results in this report are based on the synthetic project dataset and the assumptions documented above.